Your question.
Your data.

Today: one personal dataset, one verified audit.

Work independently while your instructor meets with each student. Use the Class 6 clinic hub for topic packs, verified data starting points, AI prompts and notebook cells.

Choose your starting point

I have a question + a file: Open your source and codebook. Define one row, then run the first inspection. Bring one actual output to your consultation.

I have a question, but no usable file: Name the population, comparison and needed columns. Test a small download before committing to a dataset.

I am still choosing a question: Browse a topic pack. Pick one small question and a dataset you can open today. Your first question can change after inspection.

Bring this to your consultation

My provisional question: _____

My source URL + file/version: _____

One row means: _____

Three columns I need + units: _____

My current blocker or surprising output: _____

Topic shelf

Games and player skill · Health measurements · Cybersecurity and unusual traffic · Biomedical AI and measured features · Dental care and places · Colleges, maps and majors

Original project tracks ↗ · ScholarMind AI (optional online service) ↗

A suggested source is a lead. Open its publisher page and codebook, check access terms, then verify that you can read a small file and that its variables match your question.

> **The target:** leave with your own opened dataset and an initial audit. If blocked, record exactly what is missing and agree on a recovery action.

Audit → decide → verify

Check	Record actual evidence
Raw source	File/version, collection period, population, row unit, source/terms. Preserve a raw copy.
Structure	Rows × columns, names, types, units. For media: filenames, labels and metadata.
Missingness	Count and denominator per relevant column. Check special codes in the codebook; missing is not zero.
Duplicates / keys	Full-row repeats and proposed key repeats. Are repeats valid visits, trials or frames?
Validity	Documented category/range rules, dates and units. Inspect suspicious source rows.
Verification	Hand-check three rows, reconcile counts, rerun and verify the raw file is unchanged.

My decision log

Observation + evidence: _____

Keep / flag / transform / investigate: _____

Reason + rule: _____

Rows before / affected / after: _____

What I checked myself: _____

Limitation: _____

> **No issue found?** Record the checks and keep the data unchanged. Zero parsed nulls does not prove every value, population or label is correct. Do not fill, delete or scale automatically.

For later modeling, split by the appropriate entity or time unit before fitting preprocessing parameters. Do not use the held-out test set to choose cleaning/model settings.

AI helps. You verify.

> My question is […]. One row means […]. Here are my source/codebook notes and real columns […]. Help me write one small inspection cell. Explain each line. Do not invent output or change raw data. Wait for my actual output, then suggest one independent check.

Quick checks

1. pandas reports zero missing values. What should you do next?
   My reason: _____
2. AI says “18% of prices are missing,” but has not run the code. What counts as evidence?
   My reason: _____
3. All checks you planned pass. What is a useful preprocessing result?
   My reason: _____

Answer check · after you explain

1. Check special codes, ranges and coverage; do not declare the data perfect. 2. Run the code and reconcile a hand-check; agreement between AIs is not measurement. 3. A documented “no change” is valid within the checks performed.

Save in L06 of your existing notebook

☐ Question + source record ☐ Code + actual outputs
☐ Decision log ☐ Hand-check + rerun ☐ Limitation + next action
☐ AI help used and how I verified it

Do not put private records or identifying details into an external AI chat. Use column names, documentation and an invented tiny example when needed.

Keep building the same project

6 · Audit/preprocess → 7 · Charts → 8 · Written research story + literature → 9 · First ML baseline, when appropriate → 10 · Individual next experiment.

Bring to Class 7: a rerunnable L06 audit and one question your data can answer visually.